

Project Milestone: Sentiment Classification of Restaurant Reviews

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Motivation

Online reviews play a major role in how people decide where to eat, shop or spend money. While platforms such as Yelp provide both review text and star ratings, much of today's restaurant discussion happens on social media platforms like TikTok, Instagram, etc., where people freely share their opinions but there is no formal rating system.

This project asks whether a sentiment classifier trained on Yelp reviews can generalize to unstructured social media text. By learning the relationship between review language and sentiment labels derived from Yelp ratings, we aim to build a model that can score opinion text across platforms, that can support applications such as brand monitoring and competitive analysis. The challenge is that review language is often informal, nuanced, and inconsistent, with sarcasm, mixed sentiment, and very short comments making classification difficult.

Data

We use the Yelp Open Dataset, a public collection of real-world business and review data containing 6,990,280 reviews across 150,346 businesses in 11 metropolitan areas. We work primarily from `review.json`, extracting the `text` and `stars` fields.

The input is raw review text. The target is binary sentiment. Reviews rated 4-5 stars are labeled positive, 1-2 stars negative, and 3-star reviews are dropped to reduce label ambiguity.

Data preprocessing:

To recall, our main objective is to evaluate how a model, trained on Yelp reviews from restaurants, performs in sentiment classification on TikTok comments under a business page.

The dataset – separated into businesses, users, and reviews – must be joined by us manually before we proceed with EDA. We are also focusing only on restaurant reviews for consistency. The following items describe our data preprocessing.

Keep only restaurants: Filter out businesses whose category does not include “restaurant”. Then, join all reviews from the reviews.json dataset to the filtered businesses.

Missing/invalid values: Drop records with empty text or null stars, deduplicate identical text entries.

Drop features: the review fields – funny, cool, useful – are only positive ratings on the review itself, and it does not provide helpful information about the actual restaurant itself. These fields are also *after* a review has been made.

Label mapping: Map 1 through 5 star ratings into the binary positive/negative target above. 1-2 stars = negative, and 4-5 stars = positive. We consider 3-star ratings null.

Splits: From the 100K balanced sample we use a stratified 70/15/15 split: 70,000 train / 15,000 validation / 15,000 test.

Exploratory Data Analysis (EDA):

Following will be documented from the sampled review.json:

1. Star rating distribution (bar chart): confirms the raw imbalance toward 4-5 stars.
2. Token-length vs sentiment
3. Distribution of businesses and reviews based on category

Visualization requirements:

- Average length of a review based on rating (negative vs positive)
 - Number of reviews based on rating (negative vs positive)
- Number of reviews on a given day of the week

Data challenges

Scale: ~7M reviews across businesses is not feasible to iterate on. So, we sample 100K stratified dataset.

Noise in labels: star ratings are not a very perfect reflection of sentiments. A 4-star rating might still contain complaints.

Methodology

We are planning to use 3 models with increasing complexity:

- ❖ Baseline model:
 - First use TF-IDF with Logistic regression - simple, fast, for a baseline.
 - Logistic regression learns weight for each word. Positive words ('awesome') get higher scores and negative words ('terrible') get lower scores.
- ❖ Improvement 1:
 - Next, we use TF-IDF with Linear SVM (or XGBoost). We will find out if the stronger/common model can improve performance still using the same features.
 - XGBoost builds small decision trees.
- ❖ Improvement 2:
 - Finally, we fine-tune with the Hugging Face Transformer model (like DistilBERT) which understands context and can improve classification performance/accuracy.
 - DistilBERT is a neural network model pre-trained on enormous data, which understands the context.

Contributions:

Team Member	Contribution	Artifact
Ramnik Khangura	Repo setup, Data preprocessing,	01_data_prep.ipynb
Fayed Raza	EDA and Visualizations	02_eda.ipynb
Ariv Manivasagam	Baseline models, improvements with BERT	03_baseline_models.ipynb

Martin Dela Cruz	Improvements with Neural Networks, Evaluation & Fine tuning	04_transformer_model.ipynb
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GitHub: <https://github.com/mdela/207-final-project>